

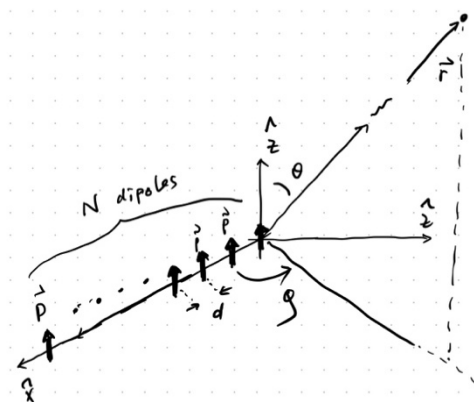
Final Exam PHYS530 (June 07, 2023)

1. (40 points) EM radiation from linear dipole arrays.

Note: The vector potential at a distance r far from ($kr \gg 1$) an electric dipole moment ($\vec{p}$)

at the origin is obtained as $\vec{A}(\vec{r}) = -\frac{i\mu_0\omega}{4\pi} \vec{p} \frac{e^{i\vec{k}\cdot\vec{r}}}{r}$, with $\vec{k} \cdot \vec{r} = kr$, here.

- a. (20) Find the general expression of the vector potential for the linear array of dipole moments as shown. *The electric dipoles are displaced with the equal separation of d along the x -direction, and linear array is made of N dipole moments.* You can take $\vec{k} = k \frac{\vec{r}}{r}$, and



distance from the second dipole to the observation point as $\vec{r}_2 = \vec{r} + d\hat{x}$ for instance, and similarly for the other dipoles.

- b. (10) Plot the angular distribution of the radiation power in the xy plane. Using $N=5$ plot the radiation power in terms of $kd \cos \varphi$.
- c. (10) Find the condition for $kd \cos \varphi$ to get the maximum as well as completely suppressed radiation power for $N=5$.

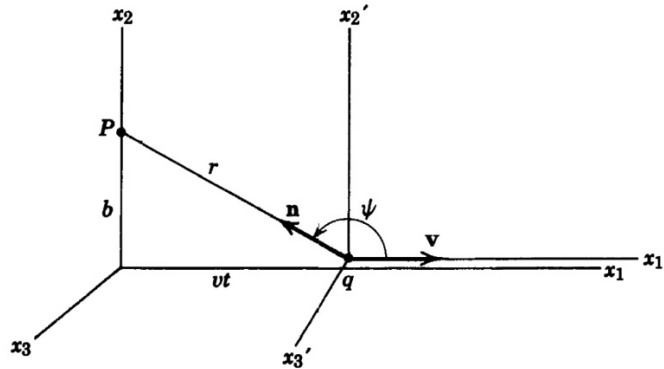
REF: Note that the angle dependent radiation power from a dipole moment is expressed

$$\text{as } \frac{dP}{d\Omega} \propto k^4 |\vec{p}|^2 \sin^2 \theta.$$

----- The final exam problem set continues in the next page -----

2. (60 points) Field tensor & fields by a moving charge

The fields at P for the moving charge as shown in the diagram below.



- (20) Find the fields $(\vec{E}', \vec{B}')$ by an observer at P moving with the charge (fields in the moving frame). The origin coincides at time zero.
- (10) Find all the components of the field tensor using $F^{\alpha\beta} = \partial^\alpha A^\beta - \partial^\beta A^\alpha$ with $A^\alpha = (\Phi, \vec{A})$.
- (10) Derive the Lorentz transformation relation below using the $F' = AF\tilde{A}$

$$\begin{aligned} E'_1 &= E_1 & B'_1 &= B_1 \\ E'_2 &= \gamma(E_2 - \beta B_3) & B'_2 &= \gamma(B_2 + \beta E_3) \\ E'_3 &= \gamma(E_3 + \beta B_2) & B'_3 &= \gamma(B_3 - \beta E_2). \end{aligned}$$

Use the Lorentz transformation matrix $A = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ for the boost

along the x_1 -axis with speed $c\beta$.

- (10) Using the Lorentz transformation relation obtained above, find out the fields at P in the rest frame.
- (10) Plot the field strength (for E_2) for fast moving charge ($\beta \sim 1$), and sketch the field lines.